EXPERIMENT NO 1

Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

Aim: -

To create an interactive diagram illustrating the components of a basic cellular system

Procedure: -

1. open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Components and Their Functions

1. ME: handset hardware identified by IMEI.
2. SIM: subscriber identity and authentication information.
3. BTS: radio transmitter/receiver for a cell.
4. BSC: controls BTSs and radio resources.
5. MSC: switches calls and manages mobility.
6. VLR: temporary subscriber information.
7. HLR: permanent subscriber profile.
8. AuC: supports authentication.
9. EIR: equipment identity status.
10. GMSC: routes calls to external networks.
11. OMC: network monitoring and maintenance.

FIGMA LINK :

https://www.figma.com/design/kzJd3S6JLy2g1PHkXLWKZZ/EXP-1?t=jRxRbwJtRBWP7omy-0

OUTPUT:

Basic Cellular System

Basic Cellular System

A cellular system divides a service area into cells. Each cell has radio equipment that connects mobile users to a wider communication network.

Circuit-switched system

A dedicated end-to-end circuit is established before a call begins. Network resources remain reserved for that call until it ends, even during pauses in speech. Traditional cellular voice calls use this approach.

Packet-switched system

Data is divided into addressed packets that share communication links with other users. No dedicated end-to-end circuit is reserved. Packets are forwarded through the network to support services such as web browsing and messaging.

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Components of a Basic Cellular System

Components of a Basic Cellular System

Mobile users communicate by radio with a base station in their cell.



The MSC switches calls, supports mobility and connects the cellular system to the public network.

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Analog Circuit-Switched System

Analog Circuit-Switched System

An analog cellular voice call uses a dedicated circuit through three main elements.

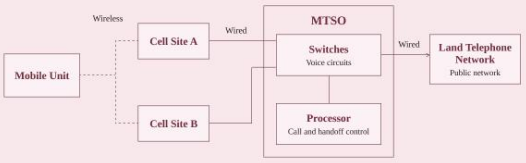
1. **Mobile Unit**
   A mobile radio terminal used by the subscriber. It transmits and receives analog voice signals over the radio link to the cell site.
2. **Cell Site**
   The radio equipment serving a cell, including antennas and transceivers. It provides coverage and carries voice and control signals to the MTSO.
3. **Mobile Telephone Switching Office (MTSO)**
   Switches calls, controls channel assignment and handoff, and interconnects the cellular system with the public land telephone network.

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Analog Circuit-Switched System Diagram

Analog Circuit-Switched System Diagram

Radio access connects the mobile unit to the wired switching network.



Dashed lines: radio links. Solid lines: wired links. The processor controls the switches.
The mobile normally uses one serving cell site; handoff transfers the call as it moves.

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Digital Circuit-Switched System

Digital Circuit-Switched System

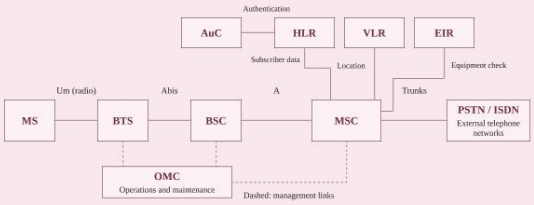
In GSM, radio access and switching functions are divided among four main elements.

1. **Mobile Station (MS)**
   The subscriber's mobile equipment and SIM; sends and receives digitally encoded voice over the radio interface.
2. **Base Transceiver Station (BTS)**
   Provides the radio transmitters, receivers and antennas that communicate with mobile stations in a cell.
3. **Base Station Controller (BSC)**
   Controls several BTSs, manages radio resources and coordinates handovers within its area.
4. **Switching Subsystem**
   Includes the MSC and associated registers. It performs call switching, mobility management, subscriber-data handling, authentication and equipment checks.

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Digital Cellular System Architecture

Digital Cellular System Architecture



HLR: Home Location Register · VLR: Visitor Location Register · AuC: Authentication Centre
EIR: Equipment Identity Register · OMC: Operations and Maintenance Centre

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Packet-Switched System

Packet-Switched System

A UMTS packet system shares network resources to carry addressed data packets.

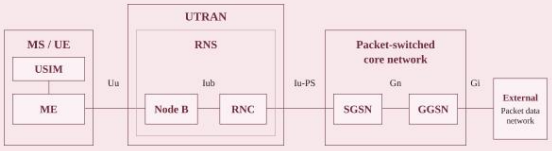
Element	Full form / name	Main function
MS	Mobile Station	The subscriber's mobile terminal.
Node B	UMTS base station	Provides the radio link; Node B is not an acronym.
RNC	Radio Network Controller	Controls Node Bs and radio resources.
SGSN	Serving GPRS Support Node	Handles mobility and packet delivery in the serving area.
GGSN	Gateway GPRS Support Node	Connects the core to external packet data networks.
CGF	Charging Gateway Function	Collects charging records for billing; not a user-data hop.

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Cellular Packet System Architecture

Cellular Packet System Architecture

UMTS user data passes from the user equipment through the radio access network to the packet core.



UE: User Equipment · USIM: Universal Subscriber Identity Module · ME: Mobile Equipment
RNS: Radio Network Subsystem · UTRAN: UMTS Terrestrial Radio Access Network

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Result: -

Hence an interactive diagram illustrating the components of a basic cellular system using Figma is created and executed successfully